For example, they combine the high-protein content of wheat with rye's high content of the amino acid lysine, which is low in wheat. It can be explained by chart below (Figure: 3.23).

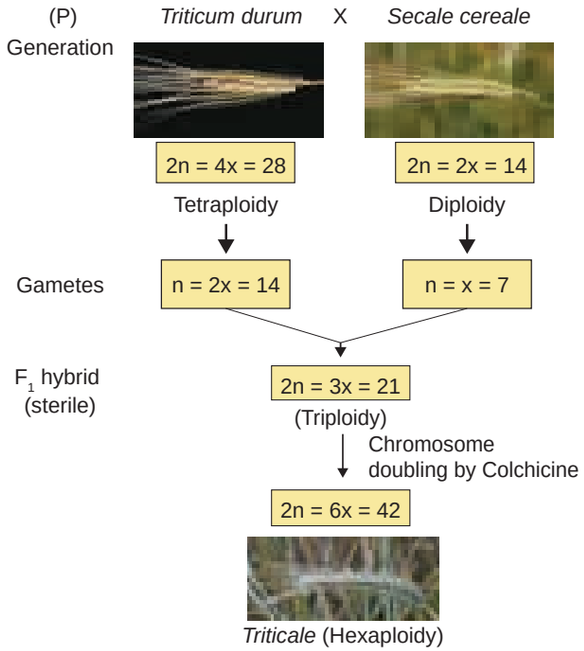
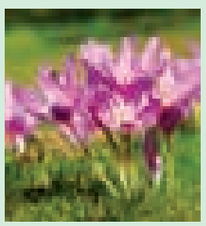


Figure 3.23 Triticale



Colchicine , an alkaloid is extracted from root and corms of ***Colchicum autumnale***, when applied in low concentration to the growing tips of the plants it will induce polyploidy. Surprisingly it does not affect the source plant *Colchicum*, due to presence of anticolchicine.

Significance of Ploidy

- Many polyploids are more vigorous and more adaptable than diploids.
- Many ornamental plants are autotetraploids and have larger flowers and longer flowering duration than diploids.
- Autopolyploids usually have higher fresh weight due to more water content.
- Aneuploids are useful to determine the phenotypic effects of loss or gain of different chromosomes.
- Many angiosperms are allopolyploids and they play a role in the evolution of plants.

Activity: Solve this

When two plants (A and B) belonging to the same genus but different species are crossed, the F_1 hybrid is viable and has more ornate flowers. Unfortunately, this hybrid is sterile and can only be propagated by vegetative cuttings. Explain the sterility of the hybrid and what would have to occur for the sterility of this hybrid to be reversed.

II Structural changes in chromosome (Structural chromosomal aberration)

Structural variations caused by addition or deletion of a part of chromosome leading to rearrangement of genes is called **structural chromosomal aberration**. It occurs due to ionizing radiation or chemical compounds. On the basis of breaks and reunion in chromosomes, there are four types of aberrations. They are classified under two groups.

A. Changes in the number of the gene loci

1. Deletion or Deficiency
2. Duplication or Repeat

B. Changes in the arrangement of gene loci

3. Inversion
4. Translocation

1. Deletion or Deficiency

Loss of a portion of chromosome is called deletion. On the basis of location of breakage on chromosome, it is divided into terminal deletion and intercalary deletion. It occurs due to chemicals, drugs and radiations. It is observed in ***Drosophila*** and Maize. (Figure 3.24)

2. Duplication or Repeat

The process of arrangement of the same order of genes repeated more than once in the same chromosome is known as **duplication.** Due to duplication some genes are present in more than two copies. It was first reported in ***Drosophila*** by Bridges (1919) and other examples are Maize and Pea**.**

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